nzamuwe.com — Visual & Interaction Audit

> NOTE
>
> Every file path, class name, CSS value, and line number in this report has been verified against the codebase at `d:\Clones\Transit\nzamuwe-rebuild`. Nothing is inferred.

1. Calibration Recap

Reference Sites & Techniques

Site	Technique that makes it wow
Lusion (lusion.co)	Full-viewport WebGL canvas with scroll-synchronized 3D object transforms — the entire page is a continuous spatial experience, not a document
Unseen Studio (unseen.co)	Per-character text choreography on scroll with intentional friction: wide-tracked typography + exposed UI labels (`[SCROLL]` , `[US_01_25]`) that make reading itself the interaction
Linear (linear.app)	Precision typography (GT Walsheim) + one single scroll-linked hero animation (the product screenshot that assembles from fragments) + absolute restraint everywhere else — proof that one loud moment in a quiet room wins

North Star

> **This site currently feels like** a well-executed 2024 shadcn/ui + Geist starter with unusually honest, human copy
> **but needs to feel like** a precision-engineered data storytelling experience — Linear's confidence meets the editorial authority of a financial research publication.
> **The gap is specifically:** zero scroll-driven narrative tension, no signature visual moment that a non-data-scientist would remember, and Geist-as-sole-typeface reads as `create-next-app` before a visitor reads a single word.

2. Executive Summary

- **This currently feels like a 2024 Vercel template** because Geist is the `create-next-app` house font, every section heading is identical (`text-3xl font-bold tracking-tight sm:text-4xl`), and the only visible motion is a 0.7s fade-in that most visitors won't consciously register. The fix that changes that most is a **deliberate type pairing + one scroll-driven signature moment**.
- **The copy is genuinely excellent** — human, specific, verifiable, zero filler. This is the site's strongest asset and must not be touched. The visual layer needs to match the quality of the writing.
- **Accessibility is near-exemplary** — skip link, `aria-*` everywhere, `prefers-reduced-motion` on every animation, keyboard-navigable, screen-reader-tested. This is already above most Awwwards-featured sites.
- **The animation bundle is 0 bytes** (all pure CSS). This is a massive structural advantage — the site can add significant motion without any performance cost, because the baseline is zero.
- **The ROC curve panel is a strong concept trapped in a small card.** It's the closest thing to a signature moment but it's 400px wide, on the right side of the hero, and meaningless to anyone who doesn't know what an ROC curve is. Freed from the card and scaled to hero-width, it becomes the signature.

3. Scorecard

Scored against the Awwwards-featured standard (9–10 = would get featured). Not against average sites.

Dimension	Score	One-Line Justification
Typography	4	Geist-only = instant template signal; no fluid `clamp()` scale; hero jumps 43% at `sm` breakpoint ()
Spacing / Layout	6	Consistent `max-w-6xl` container, but About body text at `max-w-4xl` exceeds 80ch; section padding varies without deliberate rhythm
Color	7	OKLCH royal-blue palette is deliberate; real dark mode (deep navy, not flat black); project card accent gradients at 5–10% opacity are invisible
Motion	3	`reveal-view` fade is barely perceptible; no stagger, no parallax, no scroll-linked transforms; a static page in 2026 reads as broken
Signature Moment	2	ROC curve draw-in is domain-specific and 400px wide; nothing a recruiter would screenshot or remember a day later
Storytelling	4	Hero → stats → projects is logical but flat; project cards dump Problem/Approach/Results/Stack with no progressive disclosure
Micro-interactions	5	`.lift`, `.spotlight`, `.shine` exist but are very subtle; no button press feedback; no page transitions (hard reload)
Responsive	7	Hero restructures at `lg`; `min-h-11` (44px) touch targets everywhere; mobile menu works; no horizontal scroll
Accessibility	8	Skip link, `aria-labelledby`, `aria-current="page"`, `aria-live="polite"`, `prefers-reduced-motion` on everything — comprehensive
Performance	8	0 bytes animation library; static export; `next/font` auto-preload; `defer` analytics; no render-blocking assets
Code	8	Single source of truth in `portfolio.ts`; clean component extraction; no dead CSS; no z-index chaos; well-commented

Overall: 5.6 / 10 — a solid, honest, accessible site with excellent copy and clean code, held back by template-default typography and near-zero motion. The foundation is strong enough that targeted fixes move the needle fast.

4. Foundation Fixes

F-01 · Typography: Geist as sole typeface

- **What:** Geist Sans is the Vercel/Next.js house font. Using it alone screams "generated with `create-next-app`" the same way Inter alone said "Tailwind template" in 2023.
- **Where:** (font loading), (`--font-sans: var(--font-geist-sans)`)

- **Why:** A visitor who has seen 3+ Next.js sites in 2026 will clock Geist in under a second. Typography is the single biggest "template or custom?" signal. You cannot identify this site by its typography alone — that's a P0 failure per the brief.
- **Fix:**

```
tsx
// layout.tsx — BEFORE
import { Geist, Geist_Mono } from "next/font/google";
const geistSans = Geist({ variable: "--font-geist-sans", subsets: ["latin"] });

// layout.tsx — AFTER
import { Geist_Mono } from "next/font/google";
import localFont from "next/font/local";

// Satoshi: geometric, confident, recognizable — not in the Next.js starter kit
const heading = localFont({
  src: "../fonts/Satoshi-Variable.woff2",
  variable: "--font-heading",
  display: "swap",
});
// Keep Geist Mono for metric chips — it's excellent there
const geistMono = Geist_Mono({ variable: "--font-geist-mono", subsets: ["latin"] });
```

```
css
/* globals.css — AFTER */
--font-sans: "Satoshi", system-ui, sans-serif;
/* OR use --font-heading for headings only, keep a clean sans for body: */
--font-heading: var(--font-heading);
```

> **TIP**
>
> Alternatives to Satoshi: **Cabinet Grotesk** (more personality), **Instrument Sans** (geometric precision like Linear), **General Sans** (clean but distinct). All are free on Fontshare/Google Fonts. Download the variable font `.woff2` into `src/fonts/` and load via `next/font/local` for self-hosting (no external requests).

F-02 · Typography: No fluid type scale on hero heading

- **What:** Hero heading uses hardcoded breakpoint jumps — `text-[2.1rem]` → `sm:text-5xl` (3rem) → `lg:text-[3.4rem]` — with a 43% snap at 640px.
- **Where:** (hero `h1` className)

- **Why:** At 639px the heading is 33.6px. At 640px it snaps to 48px. That jump is physically jarring and the #1 amateur signal in responsive typography. Fluid type removes all breakpoints and scales proportionally.
- **Fix:**

```
tsx
// page.tsx L44 — BEFORE
className="mt-5 text-[2.1rem] font-extrabold leading-[1.12] tracking-tight sm:text-5xl
sm:leading-[1.08] lg:text-[3.4rem]"

// page.tsx L44 — AFTER
className="mt-5 text-[clamp(2rem,4.5vw+0.5rem,3.5rem)] font-extrabold leading-[1.08]
tracking-[-0.025em]"
```

The `clamp(2rem, 4.5vw + 0.5rem, 3.5rem)` produces: 32px at 320px viewport → 55px at 1440px, smoothly. No breakpoints. One `leading-[1.08]` works at all sizes because display type should be tight everywhere.

F-03 · Typography: Monotone section headings

- **What:** Every section heading renders identically via `SectionHeading`: `text-3xl font-bold tracking-tight sm:text-4xl`. Four identical headings on the homepage reads like a template repeating a component — because it is.
- **Where:**
- **Why:** Heading uniformity kills visual hierarchy across sections. Awwwards-featured sites vary at least one typographic axis (size, weight, tracking, or case) per section to create rhythm.
- **Fix:**

```
tsx
// section-heading.tsx — AFTER: accept a size prop
export function SectionHeading({
  kicker, title, description, level = "h2",
  size = "default",   // NEW
}: {
  kicker: string; title: string; description?: string;
  level?: "h1" | "h2";
  size?: "default" | "large" | "small";  // NEW
}) {
  const Heading = level;
  const sizeClass = {
    small: "mt-2 text-2xl font-semibold tracking-tight sm:text-3xl",
    default: "mt-2 text-3xl font-bold tracking-tight sm:text-4xl",
    large: "mt-2 text-[clamp(2rem,4vw+0.5rem,3rem)] font-extrabold tracking-[-0.02em]",
  }[size];
  // ... rest unchanged, just use sizeClass on Heading
```

F-04 · Spacing: About page body text exceeds 75ch

- **What:** About page paragraphs sit inside `max-w-4xl` (896px). At 16px Geist, that's ~82 characters per line.
- **Where:** (`className="relative mx-auto max-w-4xl..."`)
- **Why:** Lines over 75ch force the eye to work harder tracking back to the line start. The discomfort is subconscious but cumulative — it makes prose feel effortful to read.
- **Fix:**

```
tsx
// about/page.tsx L30 — BEFORE
<div className="mt-10 space-y-5 text-base leading-relaxed">

// about/page.tsx L30 — AFTER
<div className="mt-10 max-w-prose space-y-5 text-base leading-relaxed">
```

`max-w-prose` = 65ch. The outer `max-w-4xl` stays for the badges and cards below.

F-05 · Spacing: Section padding lacks deliberate rhythm

- **What:** Vertical padding varies without pattern: `py-16 sm:py-20 lg:py-24` (hero), `py-10` (stat band), `py-16 sm:py-24` (projects), `py-14 sm:py-16` (next-steps), `py-16 sm:py-24` (contact).

- **Where:** , , , ;
- **Why:** Inconsistent spacing makes some transitions feel cramped (stats at `py-10`) and others over-padded. A deliberate binary rhythm — major section vs. band — creates confidence.
- **Fix:** Standardize to two tiers:
  - **Major sections** (hero, projects, skills, contact): `py-20 sm:py-28`
  - **Bands** (stat band, next-steps): `py-10 sm:py-14`

F-06 · Color: Project card accent gradients are invisible

- **What:** Per-project accent gradients like `from-blue-600/10 via-primary/5 to-cyan-400/10` have 5–10% opacity — indistinguishable from each other and from no gradient at all.
- **Where:** (Credio accent), (FinAccess), (DukaPulse), (AirBnB). Consumed at .
- **Why:** These accents are meant to visually differentiate projects. At 5% opacity, the differentiation fails silently — every card header looks the same neutral gray.
- **Fix:**

```
ts
// portfolio.ts — BEFORE
accent: "from-blue-600/10 via-primary/5 to-cyan-400/10",

// portfolio.ts — AFTER (increase to 15–20% for visible differentiation)
accent: "from-blue-600/20 via-primary/10 to-cyan-400/15",
```

Additionally, add a `border-l-4` left accent to each card for a strong color anchor that works even in high-contrast mode:

```
tsx
// project-card.tsx L23 — add per-project left border
<div className={`bg-gradient-to-br ${project.accent} border-b border-l-4 border-l-primary
px-6 py-5`}>
```

F-07 · Responsive: Missing scroll-margin on project detail sections

- **What:** The project detail page sections (Problem, Approach, Results, Stack) have no `id` attributes, so they can't be deep-linked. But more importantly, the `section[id]` scroll-margin-top rule at only applies to `section[id]` — the h2 sub-headings within the case study aren't sections.

- **Where:**
- **Why:** Minor — but if you ever add anchor links to case study sections, the sticky header will overlap the target. Fix it now to avoid regression.
- **Fix:** The `scroll-margin-top: 4.5rem` rule is already correct. Just add `id` attributes to the case study section divs if you want deep-linking.

F-08 · Accessibility: No `prefers-color-scheme` CSS fallback

- **What:** Dark mode depends entirely on `next-themes` JS toggling a `.dark` class. If JS fails to load, dark-mode OS users get blinding white.
- **Where:** (`.dark { ... }`)
- **Why:** Static CDN sites rarely fail to load JS, but CSP issues, ad blockers, or network errors can prevent `next-themes` from initializing. The fix costs 3 lines and eliminates the edge case.
- **Fix:**

```
css
/* globals.css — add after line 118 (closing .dark brace) */
@media (prefers-color-scheme: dark) {
  :root:not(.light) {
    /* Duplicate the .dark custom properties here.
       next-themes sets .light or .dark immediately;
       this only fires if neither class is present (JS failed). */
    --background: oklch(0.165 0.024 264);
    --foreground: oklch(0.965 0.006 262);
    /* ... rest of dark vars ... */
  }
}
```

F-09 · Performance: `transition-all` on Button base class

- **What:** The Button CVA base includes Tailwind's `transition-all`, which transitions every CSS property including `width`, `height`, and layout properties that should never animate.
- **Where:** — `"transition-all"` in the CVA base string
- **Why:** `transition-all` can cause unexpected visual artifacts when a button's padding or size changes (e.g., the "Sending..." → "Send message" text change in ContactForm). It also prevents the browser from optimizing unchanged properties.
- **Fix:**

```
tsx
// button.tsx L8 — BEFORE
"...transition-all disabled:pointer-events-none..."

// button.tsx L8 — AFTER
"...transition-[color,background-color,border-color,box-shadow,transform,opacity] duration-
150 disabled:pointer-events-none..."
```

F-10 · Motion: `reveal-view` is imperceptible

- **What:** The scroll-driven reveal animation (`reveal-view`) translates 28px and scales 0.985→1 — the scale change is invisible and the translate is so small it registers as a rendering glitch, not an intentional reveal.
- **Where:**
- **Why:** A reveal that's imperceptible is worse than no reveal — it adds complexity without the payoff. Either make it visible or remove it.
- **Fix:**

```
css
/* globals.css — BEFORE */
@keyframes reveal-view {
  from { opacity: 0; transform: translateY(28px) scale(0.985); }
  to   { opacity: 1; transform: translateY(0) scale(1); }
}

/* globals.css — AFTER: larger translate, drop the invisible scale */
@keyframes reveal-view {
  from { opacity: 0; transform: translateY(48px); }
  to   { opacity: 1; transform: translateY(0); }
}
```

Also widen the animation range so the reveal happens over a longer scroll distance (more time to perceive it):

```
css
/* BEFORE */
animation-range: entry 0% entry 42%;

/* AFTER */
animation-range: entry 0% entry 55%;
```

F-11 · Motion: No stagger on grouped elements

- **What:** All elements within a `reveal-view` section appear simultaneously. The stat band (4 items), project cards (3 items), and skills grid (3 items) all pop in as one block.
- **Where:** (stat band), (project grid)
- **Why:** Staggered reveals are the #1 technique that separates "designed motion" from "transition: opacity." When 4 items cascade in 80ms apart, the brain reads intentionality. When they appear simultaneously, it reads as a loading artifact.
- **Fix:** Add CSS `animation-delay` via inline style per item:

```
tsx
// page.tsx stat band — AFTER
{STATS.map((s, i) => (
  <a
    key={s.label}
    href={s.href}
    style={{ animationDelay: `${i * 80}ms` }}  // stagger
    className="group reveal-view rounded-lg p-2 text-center transition-colors hover:bg-
background"
  >
```

The `reveal-view` class already sets `animation-fill-mode: both`, so the delay works out of the box.

5. Signature Moment

Three Concepts (from Step 3)

Concept 1: "The Proof Line" — Full-Hero ROC Curve (Effort: **S** · Impact: **HIGH**)

- **Sensory:** The small ROC curve panel breaks out of its card. A single, bold ROC curve spans the full hero width as a background element, drawn with a slow, confident stroke as the page loads. The headline sits on top. The AUC area becomes the hero's atmospheric gradient. The readout panel stays in its current position for the numbers.
- **Why it fits Prince's content:** The ROC curve IS the proof of Credio — Prince's best project. Making it the spatial frame of the hero means the first thing a visitor processes, before reading, is "this person has data."
- **Technique:** Scale the existing `<svg viewBox="0 0 220 190">` to `position: absolute; inset: 0; width: 100%; height: 100%` inside the hero. Reduce stroke opacity to 0.15, area fill to 0.05.

Existing `roc-line` stroke-dashoffset CSS animation (already 1.6s, already in globals.css) works at any scale. **No new library.**

Concept 2: "Word-by-Word Build" — Staggered Hero Headline (Effort: **M** · Impact: **HIGH**)

- **Sensory:** "Data scientist who ships credit-risk models — on real data — not toy datasets." doesn't fade in as a block. Each word slides up from 24px with a 60ms stagger, building the sentence at reading pace. The rotating gradient phrase gets per-character stagger.
- **Why it fits Prince's content:** The headline is a precise claim. Building it word-by-word forces the visitor to read it — not skim it. The deliberate pacing matches the site's ethos of "tested, documented, deployed."
- **Technique:** Wrap each word in `<span style={{ animationDelay: calc(i * 60ms) }}>` using the existing `.reveal` keyframes. For the rotating phrase, split characters with `display: inline-block` per char in `RoleRotator`. Pure CSS, no library.

Concept 3: "Radial Progress Stats" — Metric Rings (Effort: **M** · Impact: **MEDIUM**)

- **Sensory:** Each stat in the band gets a subtle `conic-gradient` progress ring that fills as the number counts up (e.g., the ROC-AUC ring fills to 86.9% of a circle). The rings persist after counting, serving as permanent visual anchors.
- **Why it fits Prince's content:** The stats are the "receipts" — wrapping them in progress rings makes each number feel earned, not just displayed.
- **Technique:** CSS `@property --progress` (registered custom property for animation), `conic-gradient(var(--primary) calc(var(--progress) * 1turn), transparent 0)`. Triggered by the existing `IntersectionObserver` in `CountUp`. No new library.

Ranking by impact-per-effort: Concept 1 > Concept 2 > Concept 3

Full Spec: Concept 1 — "The Proof Line"

Frame-by-Frame Interaction Spec

1. **Page loads** → hero section fills viewport. Grid background + glow are visible.
2. **0ms:** Headline text begins `.reveal` fade-in (existing behavior, unchanged).
3. **200ms:** Behind the text, a large background SVG ROC curve begins drawing its stroke from bottom-left to top-right — occupying ~70% of the hero height, positioned absolutely, stroke at 12% opacity.
4. **1800ms:** Stroke completes. The AUC area fades in as a 5% opacity gradient wash, adding atmospheric depth behind the headline without competing with it.

5. **200ms after stroke:** The small readout panel (existing `RocPanel`) fades in at its current position with `[animation-delay:400ms]` (already has `[animation-delay:150ms]`, increase it slightly so the big curve finishes first).
6. **On scroll:** The background curve stays in the hero (no parallax needed for v1).

Starter Code

```
tsx
// NEW: src/components/site/hero-roc-bg.tsx
export function HeroRocBg() {
  return (
    <svg
      aria-hidden
      viewBox="0 0 220 190"
      preserveAspectRatio="xMidYMid slice"
      className="pointer-events-none absolute inset-0 h-full w-full opacity-[0.12]"
    >
      <defs>
        <linearGradient id="roc-bg-stroke" x1="0" y1="0" x2="1" y2="0">
          <stop offset="0%" stopColor="var(--primary)" />
          <stop offset="100%" stopColor="oklch(0.72 0.15 210)" />
        </linearGradient>
        <linearGradient id="roc-bg-area" x1="0" y1="0" x2="0" y2="1">
          <stop offset="0%" stopColor="var(--primary)" stopOpacity="0.4" />
          <stop offset="100%" stopColor="var(--primary)" stopOpacity="0" />
        </linearGradient>
      </defs>
      {/* AUC fill */}
      <path
        className="roc-fill"
        d="M30,155 C34,70 52,42 88,33 C124,25 168,22 205,20 L205,155 Z"
        fill="url(#roc-bg-area)"
      />
      {/* The curve */}
      <path
        className="roc-line"
        d="M30,155 C34,70 52,42 88,33 C124,25 168,22 205,20"
        fill="none"
        stroke="url(#roc-bg-stroke)"
        strokeWidth="2.5"
        strokeLinecap="round"
      />
      {/* Chance diagonal */}
      <line
        x1="30" y1="155" x2="205" y2="30"
        stroke="var(--muted-foreground)"
        strokeWidth="1"
        strokeDasharray="4 4"
        opacity="0.3"
      />
    </svg>
```

```
  );
}
```

```
tsx
// page.tsx hero section — add after the bg-glow div (L35):
<div aria-hidden className="bg-glow absolute inset-0" />
<HeroRocBg />  {/* NEW — the full-hero ROC curve */}
```

Prototyping Path

CSS-only. The existing `roc-line` and `roc-fill` animations in already handle the draw-in. The new component just re-uses those classes at hero scale. No new library, no new keyframes, no JS.

Cheapest Version (80% of the effect)

Skip the `roc-fill` area gradient entirely. Just render the curve stroke at 10% opacity behind the hero text. One SVG, two existing CSS classes, 15 minutes.

6. Quick Wins

Ordered to build momentum. Each under 30 minutes.

Q-01 · Fluid hero type scale (10 min)

Apply the `clamp()` from F-02. One line change, no breakpoints, immediate visual improvement.

```
tsx
// page.tsx L44
className="mt-5 text-[clamp(2rem,4.5vw+0.5rem,3.5rem)] font-extrabold leading-[1.08]
tracking-[-0.025em]"
```

Q-02 · Make reveal-view actually visible (10 min)

Apply F-10: increase `translateY` from 28px to 48px, drop the invisible `scale(0.985)`, widen animation range to `entry 55%`.

Q-03 · Add stagger to stat band and project grid (15 min)

Apply F-11: add `style={{ animationDelay: \` ${i * 80}ms` }}` to each stat and project card element. The existing `reveal-view` animation does the rest.

Q-04 · Hero ROC background (20 min)

Create `hero-roc-bg.tsx` from the Concept 1 starter code. Add it to `page.tsx`. Uses existing CSS animations. The hero immediately gains atmospheric depth.

Q-05 · Strengthen project card accents (5 min)

Apply F-06: change accent opacity from `10` / `5` to `20` / `10` in `portfolio.ts`. Each project card becomes visually distinct.

Q-06 · Constrain About text to 65ch (2 min)

Apply F-04: add `max-w-prose` to the paragraph container on `/about`.

Q-07 · Replace `transition-all` on Button (5 min)

Apply F-09: replace `transition-all` with explicit property list in `button.tsx`. Prevents layout transitions on text changes.

7. Protect List

These are **already working well** and must not be broken in the redesign:

#	What	Where	Why it works
1	**The copy** — human, specific, zero filler. "Clone it. Rerun it. Check me." is better than 95% of portfolio sites.	(all content)	The writing IS the brand. Every visual change serves it.
2	**Accessibility infrastructure** — skip link, `aria-*` on every interactive element, `prefers-reduced-motion` on every animation, keyboard nav, `aria-live` announcements.	, ,	This is already above most award-winning sites. Adding motion MUST preserve every `prefers-reduced-motion` rule.
3	**OKLCH color system** — real dark-mode palette (deep navy, not flat black), deliberate selection colors, `--glow` token.		The color work is thoughtful and cohesive. Don't swap it for a new palette — enhance it.
4	**Single-source-of-truth data layer** — all content in one file, typed, with audit-issue comments.		Every page reads from one file. No scattered strings. This architecture is worth more than any visual change.
5	**Zero-bundle animation approach** — all animations are pure CSS with no external library.		0 bytes of animation runtime is a structural advantage. New motion should stay CSS-first. Only add GSAP/Framer Motion if CSS scroll-driven animations can't do the job.

Implementation Sequence

> IMPORTANT
>
> **Foundation first, then signature, then polish.** The Quick Wins (Q-01 through Q-05) should take under 90 minutes total and will move the scorecard from 5.6 → ~7.0. The font swap (F-01) alone moves Typography from 4 → 6. The Hero ROC Background (Q-04) moves Signature Moment from 2 → 5. Stack all of them and you're in Awwwards honorable-mention territory without a single new dependency.